1i)

In the Discover phase of the Double Diamond design process, insights about the user experience are gained by identifying pain points through user feedback. I have chosen to analyse App store reviews on Duolingo, two recurring pain points I have noticed are disrupting advertising and unfair limitations in the free version.

One user expressed dissatisfaction with Duolingo about how many more ads started popping up after each lesson. The user originally loved that Duolingo let them learn a language for free instead of paying for classes, but now they felt like the constant ads after every lesson were killing their momentum.

The second pain point that I noticed was the unfair limitations in the free version of Duolingo, which limits the number of mistakes learners can make before being forced to wait. This mechanic interrupts user's learning flow and punishes them for making mistakes.

Reference:
honestragebaiter2 (2024). *Profits over education*. [App store review], 16 Nov 2024. Available at: https://apps.apple.com/us/app/duolingo-language-lessons/id570060128?see-all=reviews

Elisa_da_best (2024). *Amazing but have comments*. [App store review], 4 Aug 2024. Available at: https://apps.apple.com/us/app/duolingo-language-lessons/id570060128?see-all=reviews

1ii)

Language learners that use the free version of Duolingo need a way to progress consistently without interruptions because ads and the heart system discourage experimentation and disrupt their learning flow.

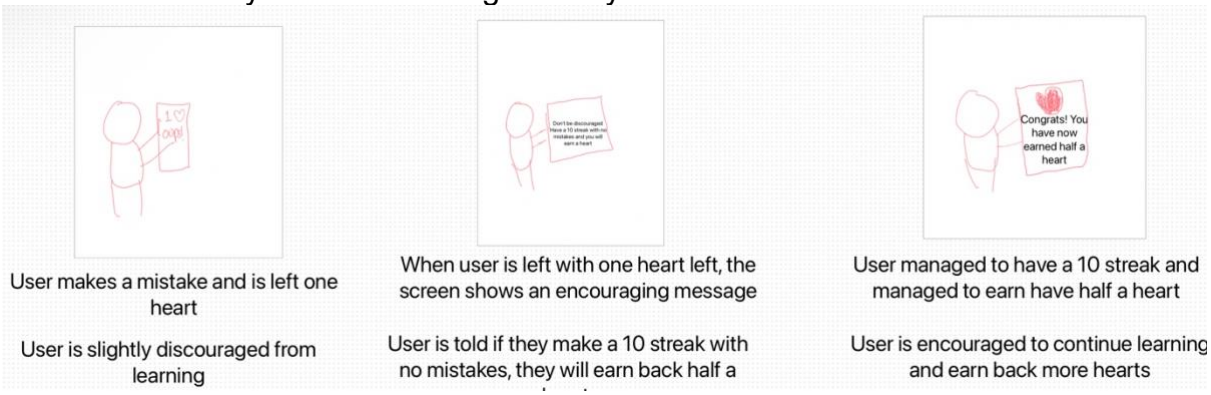
1iii)

Design concept 1:
Introduce a new "Seamless Learning Mode" for free users, where ads are grouped and shown only after every 4-6 lesson per course, instead of after each lesson. Also offer the option to view a single, skippable ad in exchange for an uninterrupted session.

This solution effectively addresses the core frustrations that free Duolingo users experience by striking a careful balance between user needs and business requirements. By reducing the frequency of ad interruptions, it allows learners to maintain their focus and build momentum during study sessions, which is crucial for language retention and confidence building.

Design concept 2:

Introduce a heart system that rewards progress rather than punish mistakes. By introducing half hearts after every 10 mistake-free streaks, users feel motivated to maintain accuracy while still having a safety net for inevitable errors.




1iv)

To evaluate the effectiveness of my solution, I would conduct a usability tests and a in-app surveys.

Usability tests would be conducted with actual Duolingo users, allowing them to experience the redesigned features firsthand and provide feedback. During these tests, focusing on how users perceive the new system as fairer and more motivating would be crucial. This would show whether the new system would encourage users rather than feel punished when making natural learning mistakes.

In-app surveys are conducted to capture the long-term impact of these changes on user behaviour and satisfaction. Key metrics to monitor include user reviews and drop-off rates when the new system is implemented, revealing whether the redesigned systems successfully retained users during these critical friction points.

1v)

One strength that Double diamonds provides is how it separates the problem space from the solution space. This helped me dive deep into the users' frustrations through reviews before starting on designing solutions. This would also encourage exploring multiple design concepts and emphasizing on analysis before implementation.

However, a limitation of the Double Diamond process is that it can be time-consuming, especially in a resource-limited environment. The Double Diamond process requires time for in-depth research, ideation, and testing, which may not always be feasible due to time constraints and tight resources.

An alternative method like Lean UX might have been more suitable in a real product team setting. Lean UX focuses on building smaller viable solutions, testing them rapidly, and adjusting based on real user feedback. This method would not only speed up the iteration cycle but also help greatly improve the product by using continuous feedback.

2i)

Norman's Seven Stages of Action model is suited for analysing onboarding experiences as it directly mirrors the user's journey from initial intent to successful completion.

User onboarding is the process of guiding new users through the application and helping them understand its features. Norman's framework focuses on error recovery analysis that helps evaluate whether the onboarding process gracefully handle user mistakes.

Norman, D (2013, The Design of Everyday Things) described a good design fundamentally includes discoverability and feedback. Onboarding flows that neglect these principles create abandonment risks for users.

Reference:
Norman, D. (2013). *The Design of Everyday Things. Massachusetts*: MIT Press.

2ii)

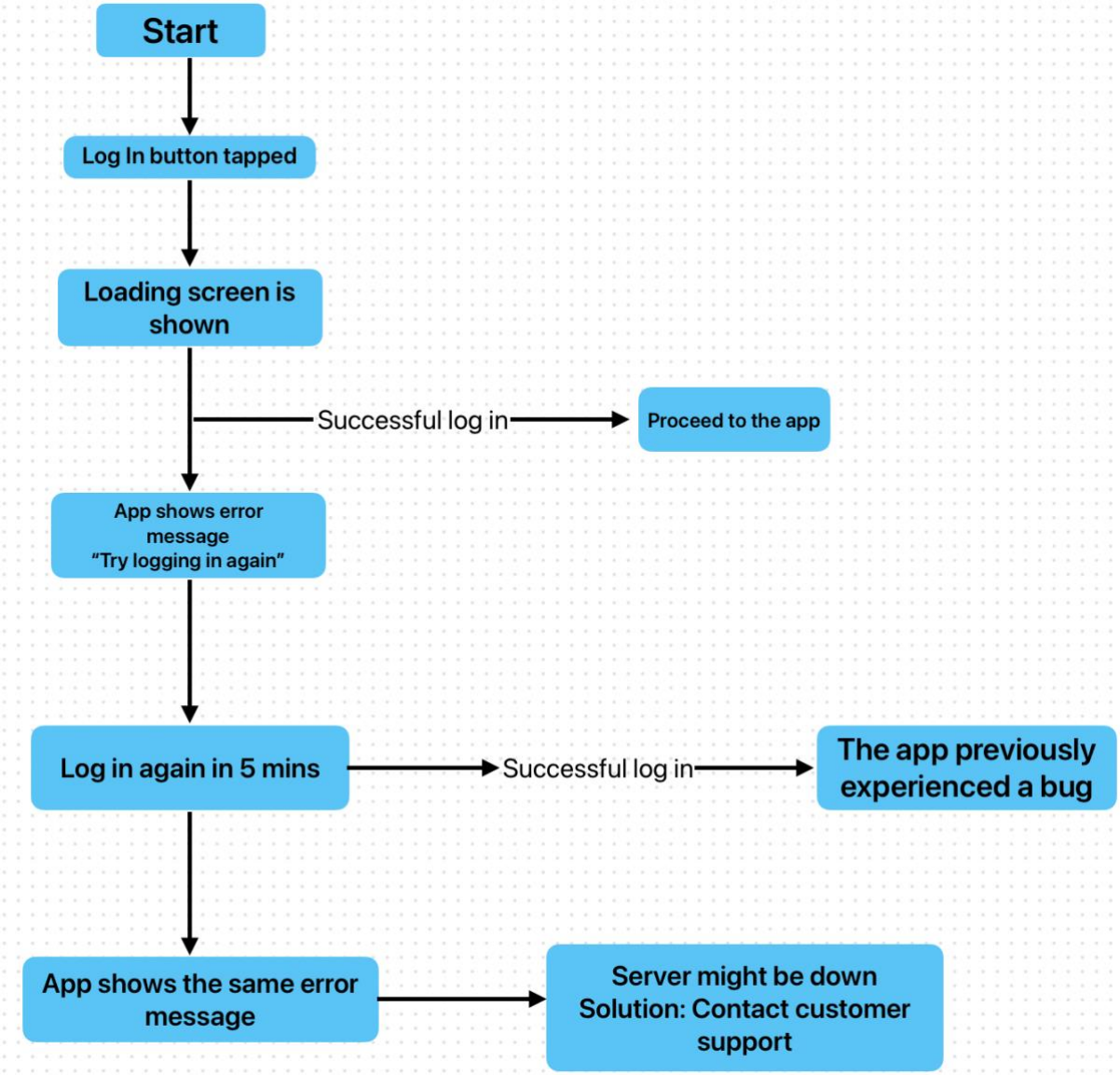
OCBC (Oversea-Chinese Banking Corporation) is a Singaporean multinational banking and financial services corporation. Its mobile banking app is widely used by Singaporeans for managing accounts, transferring funds, and paying bills.

1. Forming a goal
    - When a user logs in to the OCBC banking app, the app would block the user access without an explanation. There is no visible system status or error message provided.
2. Intention to act
    - There is a try logging in again button. Maybe pressing that button might allow the user to successfully log in. The label is vague and doesn't clarify why the previous attempt failed or what will happen next.
3. Planning the action
    - Pressing that button might reboot the app and solve the issue. The UX issue is that the system doesn't guide their expectations.
4. Executing the action
    - User taps the button and follows the log in steps.
    - The system does not guide the user's expectations
5. Feedback from action
    - App would be on the loading screen indefinitely without a timeout or error message.
    - System feedback is ambiguous with no indication of progress or failure.
6. Interpret the Feedback

- This could either mean the app is experiencing a bug or the servers could be down
- System doesn't help inform the user whether it is either a local issue or a server-side issues.

7. Evaluate the outcome
   - Try reloading the app and log in again.
   - If logging in again, try contacting support to see if the server is down

2iii)




2iv)

The reason I have chosen Norman's seven stages of action to analyse onboarding is that it effectively exposed the functional flaws in the OCBC's onboarding by mapping out user intentions to system responses and prioritising actionable solutions.

However, the model has some blind spots when it comes to analysing real-world onboarding. Firstly, the Norman model focuses on cognitive steps and would miss emotional responses from repeated failures. Lastly, the model implies a linear progression in its analysis, but onboarding is often iterative where users might use actions that are not explicitly modeled.

3i)

Primary Persona: Hannah
Demographics:
Age: 26
Occupation: Sustainability Manager
Marital status: Single
Location: Singapore
Profile: Mild mobile app user, prefers contactless payments. Often works late and does not have time to cook

Goals:

- Order healthy meals after work, mostly for dinner.
- Support sustainable brands or food chains that use eco-friendly methods.

Motivations:

- Hannah is value-driven and would prioritise vendors with discount offers.
- Loves to culturally explore dishes, would order Thai, Indian, or even Vegan meals.

Behavior Patterns: Frequency:

- Orders 4 times a week, mostly dinners
- Orders 60% plant-based or "healthier picks"; 40% multicultural treats.
- Pays by credit card for benefits

Frustrations:

- Hannah recently experienced poor Customer Service: "My vegan meal had dairy, and Uber Eats refused a refund because the rider ‘waited at the door".

3ii)

Secondary Persona: Xiao Ming

Demographics Age: 20
Occupation: University Student
Marital status: single
Location: Singapore
Profile:  Shares a flat with two other students; tight monthly allowance for food, around 300 per month.

Goals

- Get the cheapest possible meal with delivery.
- Maximize promos/discounts (e.g., 1-for-1 deals).
- Avoid cooking to save time for studies.

Motivations

- Prefers Uber Eats only during 50% off promotions, otherwise chooses to eat microwavable food.

Frustrations

- Xiao ming hates hidden fees: "The subtotal was $6.90. Why is the total suddenly $12.80 after taxes, delivery, and small order fees?"
- Limited Budget Options:  "Healthy meals cost 2x hawker food, why no affordable salads?".
- Delivery Reliability:  "My $5 McDonald’s order took 1.5 hours. Ice melted!"

Behavior Patterns

- Orders 1 to 2 times a week, sometimes only once a week.  Orders 70% fast food and 30% cheap local eats. Pays by PayNow.

3iii)

Hannah:
Demographic- Analyst notes Singapore's "younger demographics" prioritize sustainability and multicultural cuisines, aligning with urban professionals.

Goals- Analyst highlights Singapore's demand for "healthier meal options," "multiculturalism," and "sustainable practices." Hannah's goals mirror these trends.

Frustrations- The app store reviewer complained that "Uber did not even try to resolve the issue" and "Order disappeared from the app". Hannah's frustrations are direct quotes from users.

Behaviour- Analyst compares health trends to "younger demographics," suggesting frequent usage. Hannah's plant-based preference reflects the reported "surge in awareness of wellness."

Xiao Ming:
Demographic- Analyst points out an inflation in Singapore economics, impacting price-sensitive groups like students.

Goals- Analyst cites "consumer spending behavior" and "disposable income levels," explaining Xiao Ming's reliance on discounts to purchase meals.

Frustrations- The analyst's focus on "consumer spending behavior" implies cost sensitivity. Slow delivery aligns with your 2nd review's rider issues.

Behaviour- Analyst notes Singapore's "multicultural fabric" but Benny's fast-food preference contrasts Hannah's—showing how economic constraints override trends.

Reference:

cheeseburger spy (2023). -.[App store review] Available at: https://apps.apple.com/us/app/uber-eats-food-groceries/id1058959277?see-all=reviews

Statista. (n.d.). *Online Food Delivery - Singapore | Statista Market Forecast*. [online] Available at: https://www.statista.com/outlook/emo/online-food-delivery/singapore.

4i)

1. Visibility of System Status:
    - Weakness: No feedback when items become unavailable during browsing on homepage thumbnails.
2. Match between system and real world
    - Strength: Help users visualise products in real-world contexts. For example, the website displays furniture with pricing information through interactive buttons in differently styled room environments (e.g., kids' rooms, bedrooms).
    - Weakness:  It lacks to adapt to local preferences. For example, Singaporeans commonly uses PayNow as a method of payment, however IKEA priotises credit cards/Pay pal.
3. User control and freedom
    - Weakness: Lacks consistent feature parity across product variants. For example, the 3D visualisation tool is available for base product colour but absent for the other colour options.
4. Consistency and Standards
    - Weakness: Inconsistent inventory display. For example, checking availability for some items does not appear in local stores.
5. Error Prevention
    - Weakness: Lets users build large orders without warning about stock limits until checkout
6. Recognition Rather Than Recall
    - Weakness: No saved carts for logged-out users. For example, when users browse and build their shopping carts without creating an account, they will lose all their selected items in the event of closing their browser or navigating away from the site.
7. Flexibility and efficiency of use
    - Strength: Saves time by allowing users to visualize furniture before purchase by having a “view in 3D” mode
8. Aesthetic and minimalist design
    - Weakness: Overwhelming design. For example, the website displays over 50 options with furniture with no personlisation.
9. Help users recognize, diagnose and recover from errors
    - Weakness: Vague error messages. For example, screen would display "Payment process error" with no explanation or next steps when user checks out.
10. Help and documentation
    - Weakness: FAQ links buried at the bottom.

4ii)

Simulated card sorting:

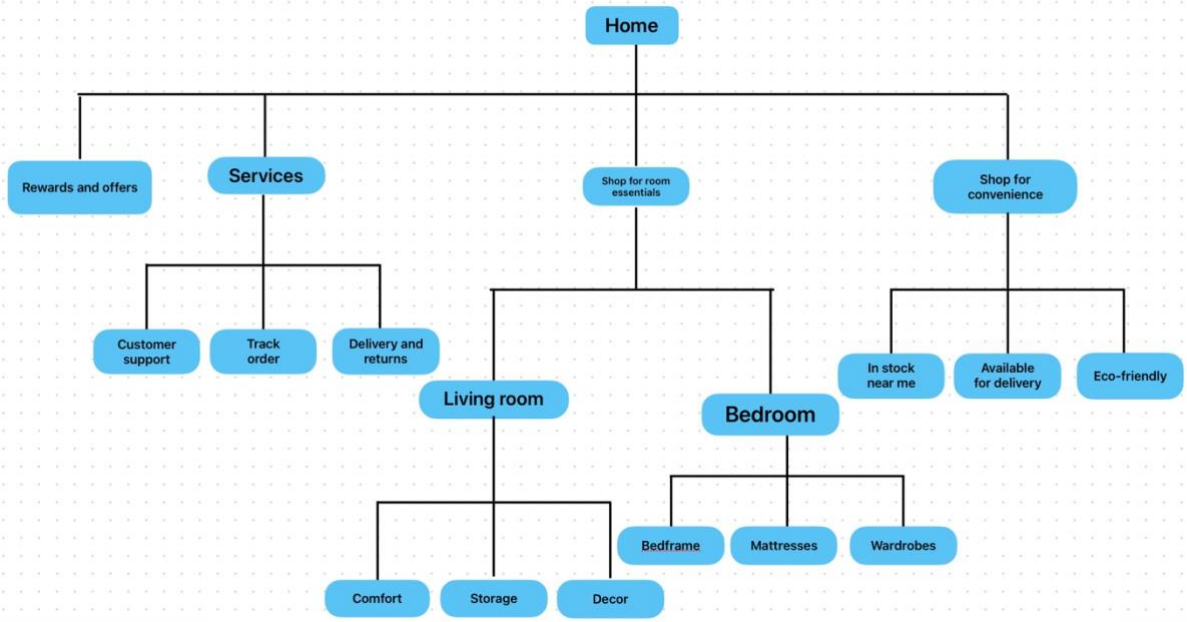
Conduct an open card sort study using a virtual platform with 20 participants. Participants were asked to organize 30 key IKEA.com items, including product categories like "Sofas" and service functions such as "Delivery Info" and "Track Order," into logical groupings that matched their mental models. The study revealed three critical issues with the current site structure.

First, users consistently struggled with overly broad category labels, particularly "Living Room," as they couldn't determine whether items like rugs belonged under room-based categories or function-based categories like "Decor."

Second, essential shopping-related services were frequently misplaced, with participants consistently grouping delivery and return information under "Shopping" rather than the current "Customer Service" placement, indicating a confusion between user expectations and the existing architecture.

Finally, users demonstrated a strong preference for inventory management tools to be prominently featured before product browsing begins, specifically requesting filters like "Show only in-stock items" to be available at the category entry point rather than buried within product listings. These findings suggest that users prioritize functional task completion over the current room-based organizational system, particularly when dealing with practical shopping concerns like availability and logistics.

Proposed IA:



4iii)

Design recommendation 1: Streamlined checkout and Error recovery

- One of the issues from my heuristic and IA findings was the vague payment errors with no support access
- One solution would be to replace “Payment error” with detailed actionable fixes. For example, the error that the user experienced with has something to do with their bank card, the screen would display “Card declined --- try another payment method or contact your bank”.
- Baymard Intstitute (2023) stated in their research that 70% of users would abandon their purchase when websites are confusing in their error messgages. Users, with no knowledge of how to proceed, would have no other option but to abandon their checkout.
  Reference:

Baymard Institute (2023). *E-Commerce Checkout Usability: An Original Research Study*. [online] Baymard Institute.
Available at: https://baymard.com/research/checkout-usability.

Design recommendation 2: Category filtering

- Another issue that was raised was the overwhelming options displayed and no pre filter for preferences (e.g stock, delivery)
- One recommendation is to add filter in the homepage that divides the broad categories. For example, there could be filter for preferences like “available for delivery” or “eco-friendly”. A broad categories could be “Living room”, and the sub-category could be “Sofas”, “Tables”, and “Lighting”
- Vassilatos, F. and et al.(2023) stated that a good filter design would allow users to feel more in control and less overwhelmed.

Reference:

Vassilatos, F. and Crawshaw, C (2023). *Filter UX Design Patterns & Best Practices - Pencil & Paper*. [online]
Available at: https://www.pencilandpaper.io/articles/ux-pattern-analysis-enterprise-filtering.

5i)

The mobile application I have chosen is the UNIQLO mobile app.

The "Forgot Password" process maintains a consistent visual style and button placement in line with the rest of the app. Form labels, buttons and color schemes are all consistent across all platforms. This aligns with the first rule

When clicking on the send password reset link button, a confirmation message was shown on the screen. After submitting the password reset form successfully, a confirmation message will be sent. This will give a clear sense of task completion which aligns with rule 4: Design Dialogs to Yield Closure

The app uses input constraints to reduce user error. For example, the app only allows email-formatted input in the email field. This reduces invalid submissions, which aligns with rule 5: offer simple error handling

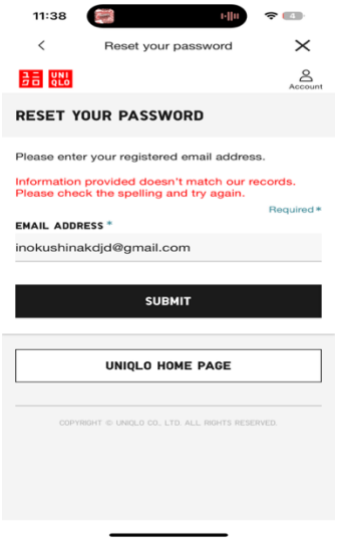
In the forget password screen, there is a “UNIQLO HOME PAGE” button that goes back to the login screen. The email input box also has a text field cursor, which allows edits of the email before submission. There is no irreversible step in this flow, which aligns with rule 6: Permit easy reversal of actions

Ther is a submit button and a home page button, which give users control when to initiate the password reset process. The interface gives clear options that support user autonomy, which aligns with rule 7: Support internal locus of control

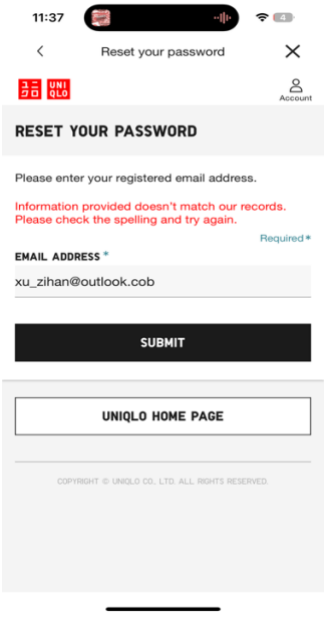
The forget password screen displays all necessary instructions and input fields, like telling the user to input their registered email into the text field. This way, users don’t need to remember information across steps which aligns with rule 8: Reduce short-term memory load

One usability issue is that error feedback is not informative enough. When entering an invalid email and a mistyped valid email account, the error feedback would be the same for both, which could be confusing for users. This violates rule 3 of the Schneiderman’s Eight Golden Rules, which is informative feedback.

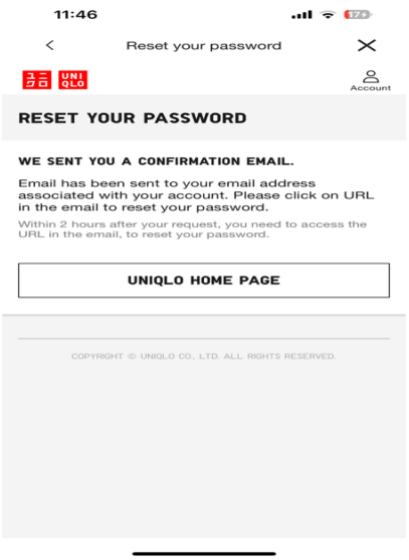
Invalid email:




Mistyped valid email:

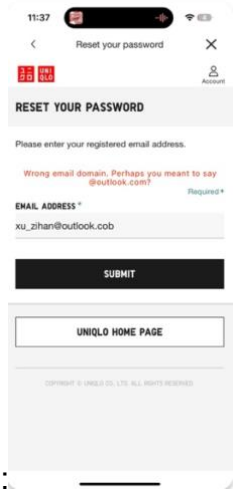



Another usability issue is that there is no “resend email” option when submitted a valid email. This violates rule 2 of the Shneiderman’s Eight Golden Rules which is the ability to allow users to use shortcuts.

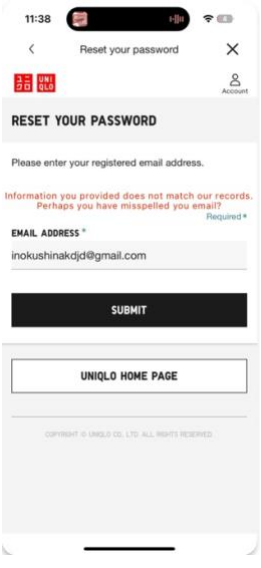



5ii)
One of the usability issues that were identified was unclear error messages, which forces users to take the time to identify the issue. According to S. Joon Park (2012): “One of the major reason for users to have frustration is due to error, as error messages tend to be inconsistent, unclear, and confusing”.

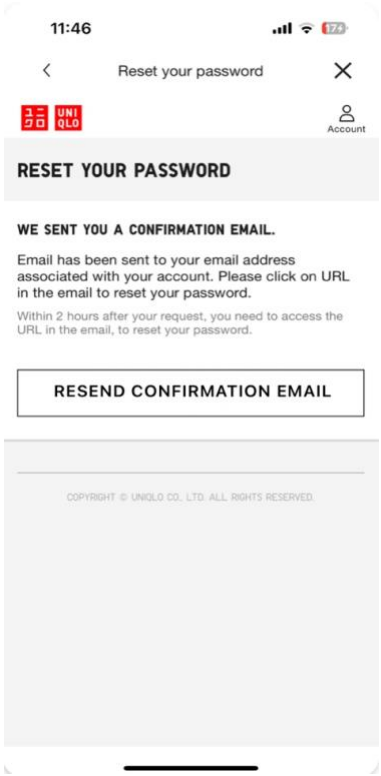
Invalid email domain:




Invalid email:




Another usability issue is that there were no resend email option, which forces the user to restart the whole process. E. Scott (2022) stated that 18.75% of users abandoned checkout due to reset email issues
Resend email redesign:




Reference:

Scott, E. (2022). *Avoid Unnecessarily Complex Password-Creation Requirements (82% Don't) – Articles*. [online] Baymard Institute. Available at: https://baymard.com/blog/password-requirements-and-password-reset.

S. Joon Park, MacDonald, C.M. and Khoo, M. (2012). *Do you care if a computer says sorry?* [online] Available at: https://dl.acm.org/doi/pdf/10.1145/2317956.2318067

5iii)
The resend email redesign aims to improve user experience by providing a an alternative when users do not receive the initial reset link. By allowing users to resend the password reset email, the system can offer better support and a shortcut so that users do not have to restart the forget password process. One limitation I have identified in the resend email redesign is that there is no alternative recovery method. My redesign lacks fallback options for users with no email access. One solution would be to send an OTP through the SMS recovery system, improving recovery rates and introduces complexity. L.Lassak et.al(2024) conducted a research that demonstrates that SMS had a higher password recovery rate (81%) than email (75%).

The redesign for invalid email and domain detection attempts to provide a more informative error message to guide users when they make mistakes. A limitation for the invalid domain name and email redesign is being overly prescriptive. When the system detects a domain error but the user actually mistyped the local part, the error message could mislead them. In *Error messages guideline*, T.Neusesser et.al. (2023) notes that overly prescriptive errors can sometimes backfire if the system misdiagnoses the issue.

Reference:

Lassak, L., Markert, P., Golla, M., Stobert, E. and Dürmuth, M. (2024). *A Comparative Long-Term Study of Fallback Authentication Schemes.* [online] Available at: doi:https://doi.org/10.1145/3613904.3642889.

Neusesser, T. and Sunwall, E. (2023). *Error Message Guidelines*. [online] Nielsen Norman Group. Available at: https://www.nngroup.com/articles/error-message-guidelines/.